

18.06 Linear Algebra - Lectures 25, 26, 27
Textbook Problem Sets - Sections 6.5, 9.2, 9.3

Problem Set Section 6.5

2) $S_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix}$

$|S_{11}| = 5 > 0$

$|S_1| = (5)(7) - (6)(6)$
 $= -1 < 0$

Fail Determinant PDM test

∴ S_1 is not positive definite.

$S_2 = \begin{bmatrix} -\frac{11}{2} & -2 \\ -2 & -5 \end{bmatrix}$

$|S_{21}| = -11/2 < 0$

$|S_2| = -1 < 0$

∴ S_2 is not positive definite.

$S_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix}$

$|S_{31}| = 1 > 0$ ✓

$|S_3| = (1)(100) - (10)(10)$
 $= 0 \leq 0$

∴ S_3 is not positive definite.

$S_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}$

$|S_{41}| = 1 > 0$ ✓

$|S_4| = (1)(101) - (10)(10)$
 $= 1 > 0$ ✓

∴ S_4 is positive definite.

$x^T S_1 x = x^T \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix} x < 0$ complete square
 $= (5)x_1^2 + 2(6)x_1x_2 + (7)x_2^2$ to
 $= 5x_1^2 + 12x_1x_2 + 7x_2^2$ get
 < 0 if $x_1 = 1$ $x_2 = 4$ $5(x_1^2 - \frac{12}{5}x_1x_2 + 7x_2^2)$
 $5(x_1 + \frac{6}{5}x_2)^2 - \frac{6}{5}x_2 + 7x_2^2$
 $5(x_1 + \frac{6}{5}x_2)^2 + (7 - \frac{6}{5})x_2^2$

$$4) f = ax^2 + 2bxy + cy^2$$

$$S_1 = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix}$$

$$\text{Let } \begin{cases} x_1 = x \\ x_2 = y \end{cases} \rightarrow S = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \text{ (✓)}$$

$$\begin{aligned} x^T S_1 x &= x^2 + 4xy + 9y^2 \\ &= (x + 2y)^2 - 2y^2 \\ &= x^2 + 4xy + 4y^2 - 2y^2 \quad \text{(✓)} \end{aligned}$$

$$X \rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Hence } f_1(x, y) = (x + 2y)^2 - 2y^2$$

$$S_2 = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix} \quad x^T S_2 x = f_2$$

$$\text{Let } x_1 = x \quad ; \quad y = x_2$$

$$\begin{aligned} x^T S_2 x &= x^2 + 6xy + 9y^2 \\ &= (x + 3y)^2 - 3y^2 \quad \text{(X)} \end{aligned}$$

$$f_2(x, y) = (x + 3y)^2 - 3y^2$$

$$\begin{aligned} f_1(x, y) &= x^2 + 4xy + 9y^2 \\ &= (x + 2y)^2 + 5y^2 \quad \text{(✓)} \end{aligned}$$

$$\begin{aligned} f_2(x, y) &= x^2 + 6xy + 9y^2 \\ &= (x + 3y)^2 \quad \text{(✓)} \end{aligned}$$

$$11) S = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}$$

$$|S|_{11} = 2 > 0 \quad \checkmark$$

$$|S|_{2 \times 2} = (2)(5) - (2)(2) = 6 > 0 \quad \checkmark$$

$$\begin{aligned} |S| &= (0)C_{31} + (3)C_{32} + (8)C_{33} \\ &= (3)[(-1)^5((2)(3) - (0)(2))] + (8)[(-1)^6(6)] \\ &= (3)(-1)(6) + (8)(1)(6) \\ &= 48 - 18 \\ &= 30 > 0 \quad \checkmark \end{aligned}$$

$$\left\{ \begin{array}{l} \text{Pivots: } p_1 = \frac{2}{1} = 2 \\ p_2 = \frac{6}{2} = 3 \\ p_3 = \frac{30}{6} = 5 \end{array} \right\} \quad \begin{array}{l} d_1 = \frac{\det A_1}{\det A_0} \rightarrow 1 \\ d_2 = \frac{\det A_2}{\det A_1} \\ d_3 = \frac{\det A_3}{\det A_2} \end{array}$$

$$\begin{aligned} 18) \quad Sx &= \lambda x \\ x^T S x &= x^T \lambda x = \lambda x^T x = \lambda \|x\|^2 \\ \lambda &= \frac{x^T S x}{x^T x} > 0 \quad \left\{ \begin{array}{l} \text{ratio of positive} \\ \text{numbers} \end{array} \right\} \\ &\Rightarrow \text{positive eigenvalues} \\ \lambda \|x\|^2 &\text{ positive eigenvalues} = \text{positive energy} \end{aligned}$$

$$19) x_i^T x_j = 0$$

symmetric matrices have
orthogonal eigenvectors

positive eigenvalues $\Rightarrow$ pos energy

20)

a) All $\lambda_s > 0$ and determinant is positive.

b) All projection matrices are singular (except I)

c) $S = LDL^T$ expresses $x^T S x$ as sum of squares
positive coefficients diagonal entries of D
are its eigenvalues

d) $S = -I$ $|S|_n = +1$ when n is even
but $\nexists S_{n \times n}$ not positive def.
as $|S_{n \times n}|_{ixi} \leq 0$

• Section 9.2 Problem Set

2, 11, 13, 14, 28

$$2) \quad u = \begin{bmatrix} 1+i \\ 1-i \\ 1+2i \end{bmatrix} \quad v = \begin{bmatrix} i \\ i \\ i \end{bmatrix}$$

$$u^H v = \begin{bmatrix} 1-i & 1+i & 1-2i \end{bmatrix} \begin{bmatrix} i \\ i \\ i \end{bmatrix} = \begin{bmatrix} & & \end{bmatrix} \quad \begin{matrix} 3 \times 1 & 1 \times 3 \\ 1 \times 3 & 3 \times 1 \end{matrix} \rightarrow$$

$$= (1-i)(i) + (1+i)(i) + (1-2i)(i)$$

$$= i - i^2 + i + i^2 + i - 2i^2$$

$$= i(1-2i)$$

$$\|u\| = \sqrt{u^H u} = \sqrt{\begin{bmatrix} 1-i & 1+i & 1-2i \end{bmatrix} \begin{bmatrix} 1-i \\ 1+i \\ 1-2i \end{bmatrix}}$$

$$= (1-i)^2 + (1+i)^2 + (1-2i)^2$$

$$= 1 - 2i + i^2 + 1 + 2i + i^2 + 1 - 4i + 4i^2$$

$$= 3 + 2i^2 - 4i + 4i^2$$

$$= 6i^2 - 4i + 3$$

$$\|u\| = \sqrt{(1-i)^2 + (1+i)^2 + (1-2i)^2}$$

$$= \sqrt{1 - 2i + 1 + 1 + 2i + 1 - 4i + 4}$$

$$= 4i \quad |1-i|^2 + |1+i|^2 + |1-2i|^2$$

$$= 2 + 2 + 5$$

$$= \sqrt{9}$$

$$= 3 \quad \checkmark$$

$$|a+bi|^2 = a^2 + b^2$$

$$\|v\| = \sqrt{v^H v} = \sqrt{(i)^2 + (i)^2 + (i)^2}$$

$$= \sqrt{3} \quad \checkmark$$

$$u^H v = \begin{bmatrix} 1-i & 1+i & 1-2i \end{bmatrix} \begin{bmatrix} i \\ i \\ i \end{bmatrix}$$

$$= (1-i)(i) + (1+i)(i) + (1-2i)(i)$$

$$= i - 1 + i + 1 + i - 2$$

$$= 3i - 2 \quad \checkmark$$

$$v^H u = \overline{u^H v} = -3i + 2 \quad \checkmark$$

$$11) Q^H Q = I$$

~~$$Q^H Q = I$$~~

$$Q^H Q = I$$

$$\therefore Q^{-1} (Q^H)^{-1} = I$$

$$\therefore Q^{-1} (Q^{-1})^H = I$$

$$\therefore Q^{-1} \text{ unitary}$$

$$QU:$$

~~$$Q^H Q = I$$~~

$$Q^H Q = I$$

$$U^H U = I$$

$$\begin{aligned} (QU)^H QU &= Q^H Q U^H U = U^H Q^H Q U \\ &= I I = U^H I U \\ &= I \end{aligned}$$

$$\therefore QU \text{ unitary}$$

$$13) A^H A = I$$

$$z^H A^H A z > 0 \text{ if } z \neq 0$$

$$\therefore z^H I z > 0$$

$$(zA)^H (Az) = \|Az\|^2$$

$$\|Az\|^2 \geq 0 \text{ if } Az \geq 0$$

~~A~~ A has independent columns

so $Az = 0$ is $z = 0$.

So if $z \neq 0$ the $\|Az\|^2 > 0$

$\therefore A^H A$ is positive definite

$$14) S = Q \Lambda Q^H$$

$$i) S_1 = \begin{bmatrix} 0 & 1-i \\ i+1 & 1 \end{bmatrix}$$

$$\lambda^2 - \lambda - (1-i)(i+1) = 0$$

$$\lambda^2 - \lambda + i + 1 + 1 - i = 0$$

$$\lambda(\lambda-1) = 0$$

$$\lambda = 0 \text{ or } \lambda = 1$$

$$\lambda^2 - \lambda + 2 = 0$$

$$(\lambda+1)(\lambda-2) = 0$$

$$\lambda = -1 \text{ or } 2$$

$$\lambda^2 - \lambda - 2 = 0$$

$$(\lambda+1)(\lambda-2) = 0$$

$$\lambda = -1 \text{ or } 2$$

$$\Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\lambda = 2:$$

$$(S_1 - \lambda I)z = 0$$

$$\begin{bmatrix} -2 & 1-i \\ i+1 & -1 \end{bmatrix} z = 0$$

$$\cancel{z_1 = 2-i}$$

$$-2z_1 + (1-i)z_2 = 0$$

$$z_1 = \frac{(1-i)z_2}{2}$$

$$\text{let } z_2 = 2$$

$$\therefore z_1 = 1-i$$

$$z = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$$

$$\lambda = -1:$$

$$\begin{bmatrix} -1 & 1-i \\ i+1 & 1 \end{bmatrix} z = 0$$

$$-z_1 + (1-i)z_2 = 0$$

$$-z_1 + (1-i)z_2 = 0$$

$$z_1 = (1-i)z_2$$

$$= (1-i)(i-1) + (1)(2)$$

$$z_1 = -(1-i)z_2$$

$$\text{set } z_2 = 1$$

$$\therefore z_1 = 1-i$$

$$= -(1-i)^2 + 2$$

$$\text{let } z_2 = 1$$

$$z_1 = i-1$$

$$z = \begin{bmatrix} i-1 \\ 1 \end{bmatrix}$$

$$z_1^H z_2 = (1-i)^H \cdot (-1+i) + (2)^H \cdot (1)$$

$$= (1+i)(-1+i) + 2(1) = 0 \checkmark$$

Orthogonal as required

for a Hermitian matrix

$$z = \begin{bmatrix} 1-i & -1+i \\ 2 & 1 \end{bmatrix}$$

$$\|z\| = \sqrt{(z^H z)}$$

$$Q =$$

$$|1-i|^2 \neq (1-i)^2$$

$$\begin{aligned} \|z_1\| &= \sqrt{|1-i|^2 + (2)^2} \\ &= \sqrt{2 + 4} \\ &= \sqrt{6} \end{aligned}$$

$$(1-i)(1+i)$$

$$\begin{aligned} \|z_2\| &= \sqrt{|-1+i|^2 + (1)^2} \\ &= \sqrt{1 + (i)^2 + 1} \\ &= \sqrt{1 + 1 + 1} \\ &= \sqrt{3} \end{aligned}$$

$$Q = \begin{bmatrix} \frac{1-i}{\sqrt{6}} & \frac{-1+i}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

~~Q~~

$$S_1 = Q \Lambda Q^H$$

$$= \begin{bmatrix} \frac{1-i}{\sqrt{6}} & \frac{-1+i}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{1-i}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ \frac{-1-i}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$(1, 1+i)$ scalar multiple of $(1-i, 2)$?

$$\frac{1}{(1-i)} = \frac{(1+i)}{(1-i)(1+i)} = \frac{1+i}{2}$$

Section 9.3 Problem Sets

1, 2, 7, 10, 11

Q: 1, 4, 5, {10-12}

$$1) \quad (3) \quad F_4 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & i \\ 0 & 1 & 0 & -i \\ 0 & 1 & 0 & -i \end{bmatrix} \begin{bmatrix} 1 & i & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & i \\ 0 & 0 & 1 & i \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & i \\ 0 & 1 & 0 & -i \\ 0 & 1 & 0 & -i \end{bmatrix}$$

$$= \begin{bmatrix} 1 & i & 1 & i \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & i & 1 & i \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & i & -1 & -i \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

(use $i^2 = -1$)

$i^2 = -1$ used in last 2 rows & columns

$$7) \quad c = (1, 0, 1, 0) = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$P: \quad c' = (c_0, c_2) = (1, 1)$$

$$c'' = (c_1, c_3) = (0, 0)$$

$$F_2 = \begin{bmatrix} 1 & 1 \\ 1 & w \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$w = e^{\left(\frac{2\pi i}{n}\right)} = e^{\frac{2\pi i}{2}} = e^{\pi i} = -1$$

$$y' = F_2 c' = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \cos(\pi) + i \sin(\pi) = -1 + i(0) = -1$$

$$y'' = F_2 c'' = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$y_j = Fc = y' + w^j y'' = y' + w^2 y'' = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$y_{j+2} = y_j' - w^j y_j'' = \begin{bmatrix} y_j' \\ 2y_j'' \end{bmatrix}$$

$$\therefore y = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} \checkmark = FC$$

$$C = (0, 1, 0, 1)$$

1. P_0 $C' = (c_0, c_2) = (0, 0)$
 $C'' = (c_1, c_3) = (1, 1)$

2. T_0 $w_2 = e^{(2\pi i / 2)} = -1$

$$F_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$y' = F_2 C' = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$y'' = F_2 C'' = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

3. C_0 $y_j = y_j' + w^j y_j''$

$$y_0 = y_0' + w^0 y_0'' = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + (-1)^0 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$y_{0+2} = y_2' + w^2 y_2'' = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + (-1)^2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$y = y_0 + y_2 = \begin{bmatrix} y_0 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

$$y = FC = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} \checkmark$$

11) Columns of Fourier matrix F are eigenvectors of cyclic permutation P .

$$PF = F\Lambda \quad P = F\Lambda F^{-1} \quad (X \rightarrow F)(X\Lambda X^{-1})$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \lambda_3 & \\ & & & \lambda_4 \end{bmatrix}$$

$$\left. \begin{aligned} \lambda_1 &= 1 \\ \lambda_2 &= i \\ \lambda_3 &= i^2 = -1 \\ \lambda_4 &= +i^3 = -i \end{aligned} \right\} \checkmark$$

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